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Anthropomorphic perception and meaning-making in human–AI interaction: a comparison of mixed reality and ChatGPT using multimodal social semiotics and LIWC

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Abstract: This study investigates how students' interactions with embodied and disembodied generative conversational AI in an English for Specific Purposes classroom indicate anthropomorphism and meaning making during interaction. Fifty ESL students interacted with two different modalities of generative AI for six weeks: an embodied conversational agent (ECA) in mixed reality (MR) and ChatGPT voice applications. The study employed multimodal social semiotics (MSS) and computational linguistic analysis (LIWC) to examine both observable multimodal conduct (gestures, gaze, and spatial orientation) and linguistic markers of social orientation. MSS findings revealed higher modal density in the MR condition, with learners demonstrating sustained gaze, forward lean, and spontaneous co-speech gestures, positioning the agent as socially accountable. LIWC analysis showed that ChatGPT interactions contained significantly higher social processes, affiliation language, and self-referential pronoun use, whereas MR interactions demonstrated higher emotional tone. The results indicate that anthropomorphism operates as both an embodied semiotic practice and a linguistically enacted social stance, shaped by interactional affordances and technological modality.

Keywords: anthropomorphism; conversational AI; mixed reality; multimodal social semiotics; LIWC; language education

1 Introduction

Since the advent of OpenAI's Large Language Models (LLMs), particularly ChatGPT, generative AI has become an integral part of contemporary educational environments, especially in language learning contexts, where interaction, feedback, and

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practice are central to pedagogical effectiveness (Bozkurt 2023). In these settings, learning is fundamentally a process of meaning-making through social interaction. LLM-based conversational AI (CAI) systems generate context-sensitive, contingent, and interactionally responsive discourse, enabling learners to engage in extended dialogues, receive adaptive feedback, and participate in scaffolded interactions resembling support traditionally provided by human tutors or coaches (Jeon and Lee 2023; Zawacki-Richter et al. 2019). As a result, these systems are increasingly positioned not merely as tools but as participants in instructional activities.

In second language learning research, CAI systems are frequently deployed as tutors, peers, or coaches (Chi et al. 2001), to scaffold performance (Liu et al. 2024), and to provide corrective or strategic feedback (Luckyardi et al. 2024). This coaching role is inherently interactive and oriented toward communicative competence. When learners interact with conversational agents, they negotiate meaning, repair misunderstandings, interpret feedback, and align their responses in real time. Systematic reviews suggest that dialogic AI systems enhance engagement, particularly when learners perceive AI as responsive and socially meaningful (Rheu et al. 2021; Zhang and Pan 2025). However, less is known about how learners construct social meaning when interacting with these systems.

A central construct linking technological design and learner meaning-making is anthropomorphism, defined as the attribution of human-like qualities, intentions, or mental states to non-human entities (Epley et al. 2007). According to the Three-Factor Theory, anthropomorphism arises from elicited agent knowledge, effectance motivation, and sociality (Epley et al. 2007, 2008). In educational AI research, anthropomorphism is associated with trust and engagement (Rheu et al. 2021). Emerging interactional research suggests that anthropomorphism is not solely a perceptual judgment but is enacted procedurally through communicative practices such as greeting, mitigation, reciprocity, and relational closing (Jindapitak and Laeha 2026). From this perspective, anthropomorphism becomes visible in the semiotic resources that learners mobilize while interacting with AI.

Interaction with AI, whether textual, voice-based, embodied, or disembodied, shapes the affordances available for meaning-making. Reviews of embodied conversational agents (ECAs) in education have reported that embodiment influences engagement, attention, and perceived social presence (Ali et al. 2024; Zhang and Pan 2025). Embodied systems may invite gaze alignment, spatial orientation, and gestures, whereas disembodied systems constrain interaction to linguistic and paralinguistic cues. However, the mechanisms through which these modalities relate to anthropomorphic engagement remain unclear, as much of the literature focuses on outcomes or satisfaction measures rather than interactional processes.

Despite its theoretical relevance, anthropomorphism in educational CAI research is typically measured using self-report instruments that assess humanness

and social presence. As Li and Suh (2021) argue, misalignment between conceptualization and operationalization produces inconsistent findings. Interaction-focused work demonstrates that learners may enact anthropomorphic scripts even when acknowledging that the AI agent is non-human (Jindapitak and Laeha 2026), suggesting that anthropomorphism operates at both reflective and procedural levels.

Recent LIWC-based research indicates that anthropomorphism may be linguistically observable through affective, social, and pronoun-based patterns, even when self-reports remain unchanged (Wood 2024). However, anthropomorphism has rarely been examined as a multimodal semiotic process integrating embodied conduct and computational linguistic markers. Meaning making during AI interaction extends beyond language to include gesture, gaze, posture, and spatial orientation, which collectively construct social accountability (Kress and van Leeuwen 2020). Few studies have combined multimodal analysis with psycholinguistic tools to examine how anthropomorphic engagement is enacted and reflected in discourse.

1.1 Research gap and purpose

Although prior research has demonstrated that anthropomorphism influences trust and engagement, educational studies overwhelmingly rely on self-report measures rather than observable interactional evidence (Rheu et al. 2021). Anthropomorphism is often treated as a design feature rather than a socially constructed process that emerges in situ (Schneider et al. 2018). While interaction-focused research shows that learners enact anthropomorphic scripts procedurally (Jindapitak and Laeha 2026) and LIWC-based studies demonstrate that linguistic markers can index anthropomorphic orientation (Wood 2024), no studies have systematically compared embodied modalities such as Mixed Reality with disembodied modalities such as the ChatGPT voice application.

Therefore, this study investigates how anthropomorphic engagement with embodied and disembodied conversational AI agents is constructed during 20-minute job interview practice sessions in a four-hour classroom. Addressing this gap requires moving beyond perceptual ratings to examine how anthropomorphic meaning is constructed in situ through multimodal interactions. Multimodal social semiotics (MSS) provides a theoretical lens for analyzing how gestures, gaze, spatial orientation, and language function together to position AI agents as socially accountable or instrumental. Complementing this qualitative semiotic analysis, linguistic inquiry and word count (LIWC) enables a systematic examination of linguistic patterns associated with social orientation and anthropomorphic engagement. By integrating MSS and LIWC, this study aligns the theoretical conceptualizations of anthropomorphism with observable multimodal and linguistic

practices, addressing conceptual, methodological, and explanatory gaps in generative AI-mediated language learning research.

However, it is important to acknowledge that these modalities differ not only in embodiment but also in accessibility and infrastructural requirements. Mixed Reality systems require specialized hardware, spatial setup, and institutional resources, whereas voice-based conversational AI can be accessed through personal mobile devices. These differences extend beyond interactional affordances and have implications for educational scalability and implementation across diverse learning contexts.

It is also worth noting that examining anthropomorphism in this study is not intended to position it as the ultimate goal. Rather, it is important to understand whether practicing job interviews with an embodied versus a disembodied conversational AI agent can help learners learn more effectively and prepare more successfully for face-to-face interviews. Job interviews are inherently situated interactions that require learners to manage relational cues, presence, and real-time communication (Slater 2009). Therefore, comparing embodied and disembodied AI agents is pedagogically relevant insofar as these modalities may differentially support situated learning and prepare learners for professional real-life scenarios.

Therefore, this study aimed to answer the following research questions:

1. What differences can be identified in learners' multimodal semiotic patterns (gesture, gaze, spatial orientation, and language) during interactions with embodied versus disembodied conversational AI coaches, and how do these patterns reflect varying degrees of anthropomorphic engagement?
2. What linguistic patterns, as measured by LIWC, distinguish interactions with embodied versus disembodied AI coaches, and how do these patterns relate to anthropomorphic orientation and meaning-making?

2 Literature review

2.1 Multimodality and anthropomorphism in AI-mediated language education

Research on AI in language education has increasingly moved beyond learner satisfaction and performance outcomes to understand how learners organize interactions with conversational systems in real time. Foundational work has been guided by the Computers as Social Actors (CASA) paradigm, which proposes that when technological systems display socially contingent cues through natural language, users apply human social norms and expectations (Nass and Moon 2000). In

educational contexts, conversational AI may therefore be treated not merely as a tool but as a socially accountable interlocutor. When AI systems respond contingently and coherently, learners orient to them as interactional partners rather than passive interfaces, organizing communicative conduct accordingly (Bozkurt 2023).

Empirical research further demonstrates that social responses to AI are linked to perceptions of agency and responsiveness. When conversational agents appear intentional, users organize interactions using interpersonal routines typically reserved for human communication. These orientations are observable not only in verbal exchanges but also in embodied conduct. Studies on multimodal conversational interactions have shown that learners coordinate spoken and embodied behaviors, including gaze, nodding, gestures, prosodic modulation, and spatial orientation, during AI interactions (André and Pelachaud 2010; Kendrick et al. 2023). Such actions accomplish greetings, mitigation, progressivity, and relational closure, often even when the AI cannot perceive them, reflecting persistent interactional norms (Gulay et al. 2025).

This perspective aligns with ecological approaches to learning, which conceptualize communication as emerging from engagement with environmental affordances (Gibson 1979; van Lier 2004). When AI produces contingent responses, learners perceive opportunities for affiliation, repair, stance taking, and reciprocity enacted through multimodal semiotic resources.

Although educational AI research frequently relies on self-reported measures of anthropomorphism and trust (Chen and Park 2021; Li and Suh 2021), such instruments capture perceptions after interaction rather than how anthropomorphism is constructed in real-time. Recent reviews have called for analytic approaches that examine observable multimodal conduct (Greilich et al. 2025). Understanding anthropomorphism as an interactional accomplishment thus shifts the focus from static perception to dynamic meaning-making in educational environments.

2.2 LIWC and anthropomorphism in conversational AI

Research on anthropomorphism in conversational AI has traditionally relied on self-report scales that assess perceived humanness, animacy, and intelligence (Bartneck et al. 2009; Salles et al. 2020). Within human–computer interaction, anthropomorphism is conceptualized as a multidimensional construct shaped by design cues, task context, and individual differences (Seeger et al. 2018). In text-based conversational agents, in which physical embodiment is absent, anthropomorphic responses are primarily elicited through linguistic behavior, perceived reasoning, empathy, and responsiveness (Greilich et al. 2025; Seeger et al. 2021). Consequently, scholars are

increasingly focusing on linguistic analysis as an indirect indicator of anthropomorphic engagement.

Abercrombie et al. (2023) argue that anthropomorphism in dialogue systems may be reflected in linguistic features, such as register, persona construction, and role alignment. Pronoun use (e.g., second-person address), affective expressions, and social behavior directed toward the agent have been identified as markers of personification (Abercrombie et al. 2023; Greulich et al. 2025).

To operationalize such indicators, several studies have employed LIWC, a dictionary-based text analysis tool that categorizes words into psychologically meaningful dimensions, such as affect, social processes, pronouns, emotional tone, and authenticity (Jin and Youn 2023; Pennebaker et al. 2015). Abercrombie et al. (2023) used LIWC categories to index anthropomorphic engagement toward conversational assistants, demonstrating how linguistic markers may serve as proxies for social orientation in human–AI interaction.

Wood (2024) advanced this approach by constructing a composite anthropomorphism score derived from LIWC categories, including authenticity, emotional tone, social behavior, affect, and pronoun use. The study found that AI literacy training reduced linguistic indicators of anthropomorphism without significantly altering self-reported perceptions, suggesting that anthropomorphism may operate at both reflective and procedural levels (Greulich et al. 2025; Wood 2024).

Despite these advances, derivation of anthropomorphic tendencies solely from linguistic data remains underdeveloped. No comprehensive validated linguistic taxonomy exists, and current approaches rely on theoretically informed category selection (Greulich et al. 2025; Wood 2024). Prior work has also focused primarily on system features that elicit anthropomorphism, rather than on extracting anthropomorphic orientation from user-generated discourse.

Finally, research indicates that perceived anthropomorphism positively predicts trust in conversational AI, although the relationship is not uniformly linear (Wood 2024). Collectively, the literature positions LIWC as a promising and evolving tool for examining anthropomorphic engagement in conversational AI contexts.

3 Theoretical framework

3.1 Multimodal social semiotics (MSS)

Multimodal social semiotics provides the primary analytical framework for examining how learners construct meaning during their interactions with conversational AI. From this perspective, communication is not reducible to linguistic exchange; rather, meaning emerges through the orchestration of multiple semiotic resources in

a social context. Spoken language, gaze, gestures, facial expressions, posture, spatial orientation, and prosody function together to accomplish social actions.

Grounded in ecological and semiotic theory (Gibson 1979; van Lier 2004), MSS conceptualizes interaction as a process in which participants perceive and act upon communicative affordances. In AI-mediated settings, affordances arise from the system's turn-taking structure, responsiveness, and apparent contingency. When learners perceive these affordances as socially meaningful, they mobilize multi-modal resources to align, mitigate, repair, and create affiliations.

Jindapitak and Laeha (2026) operationalize this perspective through the concept of multimodal action packages—bounded moments in which spoken and embodied cues jointly accomplish a single interactional function. For example, a learner's softened prosody combined with smiling and a sustained gaze may mitigate disagreement, whereas nodding and posture orientation may indicate active listening. These coordinated semiotic choices construct AI as a socially accountable interlocutor, even when the system lacks perceptual access to embodied displays.

This framework complements research on mind perception in human–AI interactions. Gray et al. (2007) argue that social engagement depends on whether users perceive the AI as possessing agency or experience. MSS extends this insight by demonstrating how such perceptions become observable in interactional conduct. Rather than treating anthropomorphism solely as a psychological attribution, MSS allows researchers to analyze how it is enacted through gaze alignment, politeness strategies, reciprocity practices, and relational closings.

By foregrounding the coordination of speech and embodied conduct, MSS enables the analysis of how embodied and disembodied AI modalities afford different interactional possibilities. An embodied system occupying perceptual space may invite sustained gaze and proxemic adjustment, whereas a disembodied voice interface may limit visible alignment while still eliciting prosodic and gestural engagement. Thus, MSS provides methodological and conceptual tools for comparing how anthropomorphic orientations are enacted across modalities.

In summary, MSS frames anthropomorphism as an interactional accomplishment rather than merely a perceptual judgment. This allows the present study to examine how learners' semiotic practices position conversational AI as a socially accountable partner or as an instrumental tool, thereby addressing gaps in prior AI education research that has relied predominantly on attitudinal measures.

3.2 LIWC and anthropomorphism

The present framework conceptualizes anthropomorphism in conversational AI as a linguistically manifested social orientation that can be operationalized via

computational text analysis. Building on the theoretical propositions of Seeger et al. (2018) and Abercrombie et al. (2023), anthropomorphism is understood not merely as an explicit belief about humanness but as an interactional stance reflected in the use of language. In text-based systems lacking embodied cues, anthropomorphic attribution is enacted through lexical choices, pronoun alignment, affective expressions, and social engagement markers (Spatola et al. 2022).

Linguistic Inquiry and Word Count (LIWC) software provides a structured method for capturing these dimensions. By mapping words to psychologically grounded categories, LIWC enables the quantification of social and affective processes embedded in user prompts. Categories such as social processes, affect, positive emotions, and second-person pronouns are theoretically aligned with anthropomorphic orientation. For instance, the frequent use of “you” to address the AI may signal the attribution of agency or interpersonal presence, whereas affective expressions (“thank you,” “appreciate”) and politeness markers may index relational framing rather than instrumental questioning (Chung and Pennebaker 2012; Kahn et al. 2007).

The composite anthropomorphism score developed in the AI literacy training study illustrates how LIWC variables can be integrated into a theoretically informed index (Wood 2024). The inclusion of authenticity and emotional tone captures a broader relational stance and communicative sincerity. When combined with explicit lexical categories, these variables provide multidimensional representations of anthropomorphic engagement.

Within this framework, anthropomorphism is treated as a patterned constellation of linguistic features, rather than a single marker. Computational linguistic analysis enables the detection of subtle interactional orientations that may not be evident in self-report data, allowing for the differentiation between reflective and enacted anthropomorphism (Cohen et al. 2021). Importantly, this framework does not assume a belief in AI sentience but interprets linguistic markers as evidence of social framing. Thus, LIWC serves as an analytical bridge between psychological theory and observable communicative practice, operationalizing anthropomorphism while remaining attentive to conceptual complexity (Kaplan and Meylani 2025).

4 Methods

4.1 Participants

This study included 50 ESL students from Hudson County Community College in New Jersey, ranging in age from 17 to 63 years ($M = 29.20$, $SD = 13.65$), with a median age of

23 years (IQR = 19–40), indicating a relatively young but demographically diverse sample. Most participants were female (76.36 %), with males representing 23.64 %. On average, participants had resided in the United States for approximately 37 months (SD = 46.23), although the length of residence varied widely (2–240 months), reflecting heterogeneous migration histories. Most participants reported not speaking English at home (87.27 %), suggesting that English primarily functions as a second language in daily life.

The sample was internationally diverse, with the largest groups originating from the Dominican Republic and Ecuador, followed by Egypt and Peru, and smaller groups from several other countries. A substantial proportion (74.55 %) had previously applied for jobs in the United States, indicating familiarity with employment-related communication contexts. In terms of self-perceived speaking confidence, most participants described themselves as neutral or somewhat confident, with smaller proportions reporting very high or very low confidence levels. While 40 % had received prior formal English training, most had not. Participants reported an average of nearly nine months of English learning ($M = 8.95$, $SD = 13.87$), with exposure ranging from one to 96 months, underscoring variability in linguistic experience within the sample.

All participants provided informed consent, and the study was approved by the Institutional Review Board. The study was part of the Skills for Success course, in which students learn career-related topics. Participants were informed that they would engage in a job interview practice session with an AI agent for 20 min every week for six weeks but were not informed about the study's specific focus on anthropomorphism and multimodal interaction patterns.

4.2 Materials

Two groups were allocated different conversational AI agents. The first group was assigned to an embodied AI agent in Mixed Reality (see Figure 1), while the second group was assigned to the ChatGPT audio program (see Figure 2). Importantly, anthropomorphic engagement was operationalized through sequentially contingent interactional behaviors (e.g., gaze shifts aligned with turn-taking, forward leans during clarification sequences, and affiliative gestures accompanying stance expressions) rather than an ergonomically necessary device posture (e.g., stabilizing a headset or holding a smartphone). This distinction follows recent multimodal conversation analysis, which differentiates a mere “embodied user–smartphone configuration” from socially oriented conduct organized through conversational sequencing (Avgustis 2024). By analyzing embodied behaviors in relation to turn structure and discursive function, and triangulating these with LIWC-based

linguistic markers, the analytic framework isolates interactional orientation from habitual device handling. Accordingly, while hardware affordances differed, the analysis distinguishes device habit from socially contingent engagement by tracking how users dynamically display their interactional engagement through gaze and body orientation, rather than assuming that their focus is entirely dictated by device handling (Avgustis 2024).

The embodied conversational AI agent was implemented using an NVIDIA large language model, Unity authoring tool, Meta's All-in-One XR SDK, and Ready Player Me. The MR group used a Meta Quest Pro head-mounted display (HMD) to interact with the AI agent. MR was utilized instead of virtual reality (VR) for various educational and practical reasons. First, MR is immersive but does not isolate learners from the real world (Alkhayat 2026). Second, unlike VR, MR is less prone to causing motion sickness or distractions (Chen and Duh 2019), allowing learners to concentrate more fully on the task at hand.

The MR agent utilized spatial audio, creating the perception that the agent's voice originated from her virtual location. The agent's gaze behavior was programmed to follow human conversational norms, including maintaining eye contact during listening, occasional brief gaze aversions during speech production, and returning the gaze when yielding the conversational floor (Argyle and Cook 1994). Body language included slight forward leans when the participant spoke, periodic head nods, and hand gestures that emphasized key points. The agent's

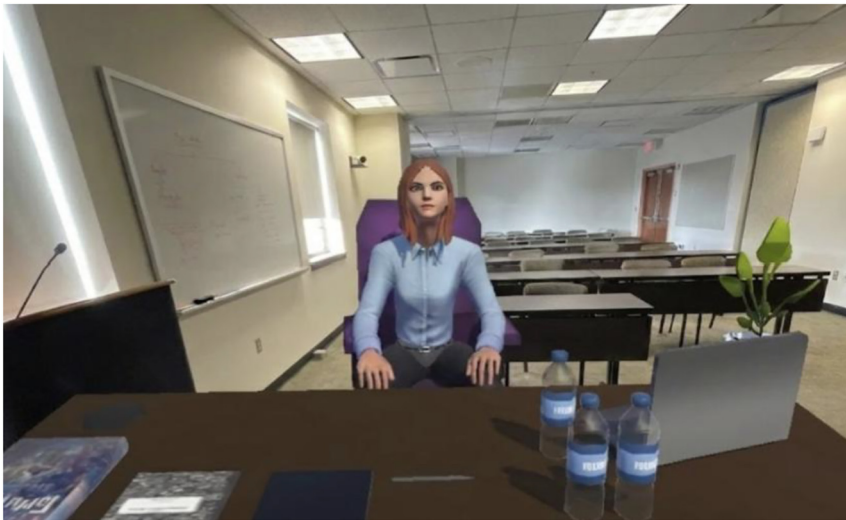


Figure 1: The embodied AI agent in mixed reality in an actual classroom.



Figure 2: ChatGPT voice mode.

conversational content was powered by GPT-4, ensuring linguistic sophistication comparable to that of the ChatGPT condition.

Participants in the ChatGPT voice condition accessed the AI agent through their smartphones. The ChatGPT voice indicated listening, signaling turn-taking, and active conversational engagement through visual and auditory cues, as shown in Figure 2.

Both agents were prompted to ask the same behavioral interview questions such as ‘Tell me about yourself,’ and ‘What is your weakness?’ The agents provided feedback on the participants’ responses, including positive reinforcement, constructive suggestions, and follow-up probing questions. The sessions lasted approximately 20 min for 6 weeks.

4.3 Data collection

The primary data for this study consisted of two types: conversation logs generated during learner interactions with the two generative AI systems (the MR-embodied conversational agent and the ChatGPT voice application) and video recordings of students while interacting with the AI agents.

Regarding the conversational logs, in terms of the MR condition, all interactional data were automatically recorded and stored within the Convai platform (convai.com), which maintains a complete transcript of user–agent exchanges. These logs included both learner input and AI-generated responses across the interaction sessions. For the ChatGPT condition, conversational logs were collected directly from the participants. Students were instructed to export or copy their ChatGPT interaction transcripts and upload them to Canvas, the institution’s learning management system (LMS), immediately after completing each session. All uploads occurred during supervised class time under the direct observation of the researcher, minimizing the possibility of post-session editing or selective alteration of conversation logs.

Although the sample included a broad age range (18–63), it was predominantly composed of younger adult learners (median age = 23). Importantly, the oldest participant was assigned to the ChatGPT condition rather than the MR group, and no observable age-related patterns emerged in emotional tone or engagement within the MR condition. Nevertheless, age-related differences in immersive technology acceptance warrant further investigation using more demographically balanced samples.

The data collection procedure ensured consistent access to complete conversational records across both modalities for subsequent linguistic and comparative analyses. The purpose of the video recordings was to examine the students’ behavior while interacting with the AI agent.

4.4 Data analysis

To examine anthropomorphic engagement across modalities, this study employed a mixed-methods analytic framework integrating multimodal social semiotics (MSS)

and computational linguistic analysis using the Linguistic Inquiry and Word Count (LIWC).

4.5 Multimodal social semiotics (MSS)

Video-recorded sessions were analyzed using an interaction-focused micro-skill sequence (MSS) procedure adapted from Jindapitak and Laeha (2026). The recordings were first reviewed holistically and then segmented into time-stamped multimodal action units, defined as bounded moments in which speech, gestures, gaze orientation (operationalized via head direction), and spatial positioning co-occurred to accomplish a single interactional function.

Each participant's six sessions were coded, and behavioral indicators were aggregated at the participant level to preserve the independence of observations ($N = 25$ per condition). Therefore, the primary unit of analysis for inferential comparison was the participant and not the individual action units.

The following observable behaviors were coded dichotomously (present at least once vs. not present across sessions):

- Spontaneous co-speech gesture
- Forward lean during clarification or repair
- Sustained head/torso orientation toward the agent/device
- Spatial interaction attempts

The proportion of participants exhibiting each behavior was compared between modalities using chi-square tests of independence. The effect sizes were calculated using phi (ϕ).

Multimodal behaviors were coded dichotomously (present vs. not present across sessions) to determine whether participants incorporated specific semiotic resources into their interactional repertoire during sustained engagement. The analytic focus was not on behavioral intensity or density, but on whether embodied resources became available as part of learners' meaning-making practices over six weeks. Given the length and multimodal scope of the dataset, this approach enabled reliable cross-participant comparisons while minimizing the inflation of minor or isolated behaviors.

To ensure reliability, a second trained researcher independently coded 20 % of the dataset using the same framework. Inter-rater reliability was calculated using Cohen's kappa for each code category. Kappa values ranged from 0.74 to 0.81, indicating substantial to almost perfect agreement (Landis and Koch 1977). Discrepancies were resolved through analytic consensus before coding the full dataset.

4.6 LIWC linguistic analysis

Linguistic analysis was conducted using the LIWC2015 program (Pennebaker et al. 2015). All conversational transcripts were cleaned and speaker-separated before analysis. Crucially, only learner-generated utterances were included in the LIWC analysis, and AI-generated turns were removed to ensure that the linguistic measures reflected the participants' anthropomorphic orientation rather than the system output. The transcripts were manually cross-referenced with the video recordings of the sessions to resolve any phonetic ambiguities and to ensure that non-native disfluencies did not result in data loss. This normalization process ensures lexical accuracy for LIWC, which research shows is a robust tool for analyzing ESL corpora as its 'bag-of-words' methodology identifies psychological markers regardless of non-standard syntax (Pennebaker et al. 2015).

Each participant completed six 20-minute sessions over six weeks. For statistical modeling, transcripts were aggregated per participant across sessions, and LIWC category scores were computed at the participant level as percentage-based metrics. Thus, the statistical unit of analysis was the participant ($N = 25$ per condition), ensuring the independence of observations.

4.7 Selection of anthropomorphism-relevant variables

Based on prior literature linking anthropomorphism to agency attribution, social cognition, and relational processing (Epley et al. 2007; Seeger et al. 2018), the following LIWC categories were selected: Social Processes, Personal Pronouns (including first- and second-person forms), Affective Processes, Affiliation (drive), Perceptual Processes, Cognitive Processes, and the summary variables Analytic Thinking, Clout, Authenticity, and Emotional Tone. These variables were selected because anthropomorphism involves attributing agency, intentionality, emotion, and social presence to non-human systems.

4.8 Assumption testing

Prior to the inferential analysis, the statistical assumptions were evaluated. The normality of the participant-level LIWC scores was assessed using Shapiro–Wilk tests and Q–Q plot inspection. Homogeneity of variance was tested using Levene's test. Welch's t -test was applied when the homogeneity assumptions were violated. All tests were two-tailed, with $\alpha = 0.05$.

4.9 Inferential analyses

Independent samples *t*-tests were conducted to compare participant-level LIWC scores between the ChatGPT and mixed reality conditions. Given the multiple comparisons across the LIWC categories, the Benjamini–Hochberg false discovery rate (FDR) correction was applied to control for Type I error while maintaining statistical power.

Effect sizes were calculated using Cohen's *d* and interpreted using conventional thresholds (0.20 = small, 0.50 = medium, 0.80 = large). All reported statistics included means, standard deviations, *t*-values, degrees of freedom, *p*-values, FDR-adjusted *p*-values, and effect sizes.

5 Results

Each of the 25 participants per condition engaged in 120 min of AI-mediated interaction over six weeks, yielding approximately 3,000 min of video-recorded data per group. These recordings were synchronized with full conversational logs, resulting in a large multimodal corpus encompassing verbal output, gestures, gaze, and spatial positioning. The volume of data allowed for both fine-grained multimodal analysis and computational linguistic comparisons.

For the LIWC analysis, a balanced corpus of 150,000 words per condition was extracted from a larger dataset. Equal sampling was employed to ensure comparability across modalities and to prevent differences in the total word count from biasing percentage-based LIWC outputs. Because LIWC generates relative frequency scores rather than raw counts, using equivalent corpus sizes strengthens statistical validity and interpretability across groups.

5.1 Multimodal behavioral patterns

To examine differences in multimodal meaning-making across modalities, participant-level behavioral indicators were compared between the mixed reality (MR) and ChatGPT voice conditions (*N* = 25 per group). Proportions were compared using the chi-square test of independence.

Spontaneous co-speech gestures were observed in 84 % (21/25) of the participants in the MR condition compared with 32 % (8/25) in the ChatGPT condition (Figure 3). This difference was statistically significant, $\chi^2(1, N = 50) = 14.88$, $p < 0.001$, $\phi = 0.55$, indicating a large association between interaction modality and the likelihood of



Figure 3: A student from the MR group practicing job interview questions with a conversational AI agent in the MR.

gesture production. Importantly, nearly one-third of the participants in the voice-only condition produced gestures in the absence of a visible interlocutor.

Forward body lean in moments of clarification or apparent misunderstanding occurred in 68 % (17/25) of the MR participants and 16 % (4/25) of the ChatGPT participants, $\chi^2(1, N = 50) = 13.52$, $p < 0.001$, $\phi = 0.52$. Sustained head and torso orientation toward the conversational agent/device during AI speech was observed in 72 % (18/25) of the MR participants compared to 40 % (10/25) of the ChatGPT participants ($\chi^2(1, N = 50) = 5.12$, $p = 0.024$, $\phi = 0.32$).

In the MR condition, two participants (8 %) attempted to physically interact with virtual objects in the environment (e.g., reaching toward virtual books or objects), reflecting their engagement with shared spatial affordances. No comparable spatial interaction attempts were observed under ChatGPT conditions.

By contrast, the ChatGPT condition exhibited patterns consistent with lower-level multimodal orchestration. Sixty percent (15/25) of the participants placed their phones flat on the table, maintaining a downward gaze orientation (Figure 4). Twenty-eight percent (7/25) held their phones in one hand with minimal gestural accompaniment. Additionally, 44 % (11/25) intermittently looked away from the device during pauses, and 36 % (9/25) moved their ears closer to the phone while



Figure 4: A student from the ChatGPT group practicing job interview questions with ChatGPT.

lowering their voices, suggesting uncertainty regarding microphone sensitivity or turn-taking timing.

In the ChatGPT condition, postural configurations were predominantly neutral or closed (e.g., inward torso orientation, limited arm movement, and minimal proxemic adjustment), whereas the MR condition was characterized by more frequent forward lean, sustained orientation, and gesture–speech coordination.

In summary, the MR condition demonstrated significantly higher rates of gesture use, clarification-oriented forward lean, and sustained orientation toward the conversational partner, indicating greater multimodal density in the interaction. The ChatGPT condition, while linguistically interactive, exhibited reduced embodied coordination and fewer spatial alignment behaviors.

5.2 LIWC linguistic analysis

The study sample consisted of 50 students, with 25 participants assigned to the ChatGPT voice condition and 25 assigned to the mixed reality (MR) conversational agent condition. Each group comprised 150, 000 words.

All LIWC variables were treated as continuous percentage scores. Independent samples *t*-tests with Welch’s correction (when appropriate) were conducted to compare linguistic markers across conditions. To control for Type I error due to multiple comparisons, the Benjamini–Hochberg false discovery rate (FDR) procedure was applied. Descriptive statistics and inferential results are presented in Table 1.

Table 2 presents participant-level LIWC comparisons between the ChatGPT voice and MR conditions. After applying the Benjamini–Hochberg false discovery rate (FDR) correction, six variables remained statistically significant.

Emotional tone was significantly higher in the MR condition than in the ChatGPT condition ($t(48) = -3.51$, FDR-adjusted $p = 0.003$, $d = -0.99$), indicating that MR interactions were characterized by more positive emotional expression. The effect size was large in this study.

In contrast, participants in the ChatGPT condition demonstrated significantly higher social process language, $t(48) = 5.97$, FDR-adjusted $p < 0.001$, $d = 1.69$, as well as higher affiliation language, $t(48) = 3.21$, FDR-adjusted $p = 0.004$, $d = 0.91$. ChatGPT interactions also showed greater overall pronoun use, $t(48) = 2.50$, FDR-adjusted $p = 0.026$, $d = 0.71$, higher Personal Pronoun use (ppron), $t(48) = 3.52$, FDR-adjusted $p = 0.003$, $d = 1.00$, and greater First-Person Singular (“I”) usage, $t(48) = 4.13$, FDR-

Table 1: Means, standard deviations, and independent samples *t*-tests comparing ChatGPT and mixed reality conditions.

Variable	ChatGPT M (SD)	MR M (SD)	<i>t</i> (48)	<i>p</i>	Cohen’s <i>d</i>
Analytic	42.55 (11.98)	45.62 (10.59)	−0.96	0.342	−0.27
Clout	78.67 (10.62)	82.91 (7.60)	−1.62	0.112	−0.46
Authentic	45.14 (17.71)	49.81 (12.76)	−1.07	0.291	−0.30
Tone	92.56 (6.00)	97.01 (2.08)	−3.51	0.001	−0.99
Social	20.52 (2.09)	17.38 (1.59)	5.97	<0.001	1.69
Affect	7.07 (1.15)	7.44 (0.88)	−1.31	0.198	−0.37
Affiliation	2.58 (0.60)	2.07 (0.52)	3.21	0.002	0.91
Pronoun	16.57 (2.01)	15.14 (2.06)	2.50	0.016	0.71
Personal Pronoun (ppron)	12.99 (2.05)	11.14 (1.66)	3.52	0.001	1.00
First-Person Singular (“I”)	5.81 (1.67)	4.04 (1.35)	4.13	<0.001	1.17
Second-Person (“You”)	6.09 (1.06)	6.41 (1.03)	−1.10	0.276	−0.31

Table 2: LIWC comparisons between ChatGPT voice and mixed reality conditions (participant-level means, $N = 25$ per group).

Variable	ChatGPT M (SD)	MR M (SD)	$t(48)$	p	FDR-adjusted p	Cohen's d
Analytic	42.55 (11.98)	45.62 (10.59)	−0.96	0.342	0.377	−0.27
Clout	78.67 (10.62)	82.91 (7.60)	−1.62	0.112	0.141	−0.46
Authentic	45.14 (17.71)	49.81 (12.76)	−1.07	0.291	0.356	−0.30
Emotional Tone	92.56 (6.00)	97.01 (2.08)	−3.51	0.001	0.003	−0.99
Social Processes	20.52 (2.09)	17.38 (1.59)	5.97	<0.001	<0.001	1.69
Affect	7.07 (1.15)	7.44 (0.88)	−1.31	0.198	0.238	−0.37
Affiliation	2.58 (0.60)	2.07 (0.52)	3.21	0.002	0.004	0.91
Pronoun	16.57 (2.01)	15.14 (2.06)	2.50	0.016	0.026	0.71
Personal Pronoun (ppron)	12.99 (2.05)	11.14 (1.66)	3.52	0.001	0.003	1.00
First-Person Singular (“I”)	5.81 (1.67)	4.04 (1.35)	4.13	<0.001	<0.001	1.17
Second-Person (“You”)	6.09 (1.06)	6.41 (1.03)	−1.10	0.276	0.331	−0.31

Note. FDR-adjusted p -values were calculated using the Benjamini–Hochberg procedure ($q = 0.05$). Significant effects after FDR correction appear in bold (Emotional Tone, Social Processes, Affiliation, Pronoun, ppron, First-Person “I”).

adjusted $p < 0.001$, $d = 1.17$. These effects ranged from moderate to very large. No statistically significant differences were observed for Analytic Thinking, Clout, Authenticity, Affect, or Second-Person pronoun use after FDR correction.

The linguistic data indicate that ChatGPT interactions were characterized by greater social and self-referential language, whereas MR interactions were marked by a higher emotional tone. In the ChatGPT condition, students frequently produced explicit verbal alignment and affiliative markers. For example, participants used stance tokens such as “Yeah,” “I see,” and “Right,” as well as affiliative evaluations including “That’s very, very, very nice” and “That’s exactly what I needed.” These utterances functioned as overt signals of agreement, acknowledgment, and relational stance, contributing to significantly higher scores in the LIWC categories of Social Processes ($d = 1.69$) and Affiliation ($d = 0.91$).

By contrast, in the MR transcripts students frequently used affectively and relationally marked expressions such as “Oh great, thank you for helping me,” “this interview practice is very important,” and “See you next time.” These utterances contributed to higher *Emotional Tone* values through the presence of positive affect terms, intensifiers, and affiliative closings, reflecting a more explicitly affect-laden interactional style.

Table 3 presents participant-level comparisons of multimodal behaviors coded through the MSS analysis. Chi-square tests revealed statistically significant differences among all four behavioral indicators.

Table 3: MSS behavioral comparison across modalities ($N = 25$ per group).

Behavior	MR % (n)	ChatGPT % (n)	$\chi^2(1)$	p
Spontaneous co-speech gesture	84 % (21)	32 % (8)	14.88	<0.001
Forward lean for clarification	68 % (17)	16 % (4)	13.52	<0.001
Sustained orientation toward agent/device	72 % (18)	40 % (10)	5.12	0.024
Device/agent gaze maintained during AI speech	76 % (19)	44 % (11)	5.41	0.020

Note. Chi-square tests were used to compare the proportion of participants exhibiting behavior at least once during the coded sessions.

Spontaneous co-speech gestures occurred significantly more frequently in the MR condition (84 %) than in the ChatGPT condition (32 %), $\chi^2(1) = 14.88$, $p < 0.001$. Similarly, forward lean during clarification was more prevalent in MR (68 %) than in ChatGPT (16 %), $\chi^2(1) = 13.52$, $p < 0.001$.

Sustained orientation toward the conversational partner was also significantly higher in the MR condition (72 %) than in the ChatGPT condition (40 %) ($\chi^2(1) = 5.12$, $p = 0.024$). Likewise, maintenance of gaze orientation during AI speech was observed more frequently in MR (76 %) than in ChatGPT (44 %), $\chi^2(1) = 5.41$, $p = 0.020$.

These results indicate that participants in the MR condition exhibited significantly greater embodied alignment behaviors, including gesture coordination, spatial orientation, and clarification-related posture shifts, than participants interacting with the ChatGPT voice. In contrast, the ChatGPT condition demonstrated reduced multimodal density, consistent with a more device-centered interactional configuration.

Tables 2 and 3 demonstrate that modality shaped both linguistic and embodied indicators of anthropomorphic engagement, with MR eliciting stronger embodied alignment and ChatGPT eliciting a stronger social-linguistic orientation.

5.3 Differences between conditions

Significant group differences emerged for some variables. Emotional Tone was significantly higher in the MR condition ($M = 97.01$, $SD = 2.08$) than in the ChatGPT condition ($M = 92.56$, $SD = 6.00$), $t(48) = -3.51$, $p = 0.001$, $d = -0.99$.

In contrast, participants in the ChatGPT condition demonstrated significantly greater use of Social Processes ($M = 20.52$, $SD = 2.09$) than those in the MR condition ($M = 17.38$, $SD = 1.59$), $t(48) = 5.97$, $p < 0.001$, $d = 1.69$. Similarly, Affiliation language was higher in the ChatGPT condition ($M = 2.58$, $SD = 0.60$) than in the MR condition ($M = 2.07$, $SD = 0.52$), $t(48) = 3.21$, $p = 0.002$, $d = 0.91$.

Pronoun usage also differed significantly between the conditions. The ChatGPT group exhibited higher overall pronoun use, $t(48) = 2.50$, $p = 0.016$, $d = 0.71$, higher personal pronoun use (ppron), $t(48) = 3.52$, $p = 0.001$, $d = 1.00$, and greater first-person singular (“I”) usage, $t(48) = 4.13$, $p < 0.001$, $d = 1.17$. The effect sizes for these differences ranged from medium to very large.

No statistically significant differences were found between the ChatGPT and mixed reality conditions for analytic thinking, $t(48) = -0.96$, $p = 0.342$; clout, $t(48) = -1.62$, $p = 0.112$; authenticity, $t(48) = -1.07$, $p = 0.291$; affect, $t(48) = -1.31$, $p = 0.198$; or second-person pronoun use (“you”), $t(48) = -1.10$, $p = 0.276$.

After applying the Benjamini–Hochberg FDR correction, the differences in Emotional Tone, Social Processes, Affiliation, overall pronoun use, personal pronouns, and first-person singular (“I”) remained statistically significant, indicating that these effects were robust to adjustment for multiple testing.

The interaction modality significantly influenced the linguistic patterns associated with anthropomorphic engagement. Participants interacting with ChatGPT’s voice exhibited greater social language use, higher affiliation-related language, and more self-referential pronoun usage, whereas participants in the mixed reality condition demonstrated a significantly higher emotional tone. These findings suggest that the interaction modality shapes how users linguistically construct and relate to conversational agents.

6 Discussion

The findings indicate that interactional patterns differed systematically across modalities in terms of how learners constructed anthropomorphism during AI-mediated coaching sessions. Consistent with social semiotic theory, anthropomorphism is not confined to verbal exchange but emerges through coordinated multi-modal resources (Kress 2009). Learners in the mixed reality (MR) condition exhibited a higher modal density, characterized by the integration of gaze orientation, gestures, spatial alignment, and speech. Sustained orientation toward the avatar, forward lean during clarification, and spontaneous co-speech gestures were more frequently observed in this condition, suggesting a pattern of embodied alignment consistent with treating the agent as an interactionally accountable presence.

Conversely, the elevated scores for ‘Social Processes’ and ‘Affiliation’ in the ChatGPT condition suggest a modality-specific compensatory mechanism (Boyle et al. 1994). In the absence of the non-verbal cues present in the MR environment, such as nodding, smiling, or eye contact, students shifted their social-relational intent entirely into the verbal channel to overcome what Cyr et al. (2007) identify as the ‘impersonal and automated’ nature of disembodied interfaces. Qualitative evidence

from the transcripts indicates an ‘over-narration’ strategy, where students employed redundant polite intensifiers (e.g., ‘That’s very, very, very nice’) and frequent stance tokens (e.g., ‘I see,’ ‘Right’) to maintain rapport. This strategic linguistic adaptation suggests that while MR elicits a visceral, embodied response to a physical presence, voice-only agents like ChatGPT necessitate a high density of explicit affiliative markers to bridge the ‘embodiment gap’ (Cyr et al. 2007) and artificially generate the sense of human warmth and sociability required for a productive social connection.

However, these interaction patterns must be contextualized within the users’ level of technical proficiency. Because our participants were not experienced MR users, the results may have been influenced by a novelty effect. The novelty effect describes an increase in motivation or engagement driven primarily by the newness of technology (Koch et al. 2018; Lee et al. 2025).

Although the novelty of the medium is a factor, the specific physical configurations of each device appeared to exert a more direct influence on user behavior. The higher prevalence of gestures in the MR condition suggests that embodied affordances may invite more multimodal interactions during AI-mediated interactions. Wearing a head-mounted display and orienting toward a spatially situated avatar likely supported the integration of gaze, gestures, and speech. However, the presence of spontaneous cospeech gestures in 32 % of the ChatGPT-voice participants is theoretically notable. Prior research has shown that gestures can accompany speech even when partners cannot see the speaker, including in mobile phone and blind-speaker contexts (Iverson and Goldin-Meadow 1998). These findings indicate that gestures can be closely coupled with speech production rather than being dependent on visual co-presence. Thus, embodiment may amplify multimodal density; however, dialogic interaction itself appears sufficient to activate conversational schemas typically associated with human interlocutors.

In contrast, ChatGPT voice interactions are characterized by a comparatively reduced multimodal orchestration. Participants more often positioned the device laterally, adopted closed postures, or adjusted their vocal volume in ways consistent with device-oriented interactions. Although some participants verbally described the agent in relational terms (e.g., “like an older brother”), such descriptions did not consistently correspond with embodied alignment behaviors. This divergence aligns with interactional research suggesting that anthropomorphic attribution may occur at the cognitive or verbal level without being fully enacted through multimodal conduct.

The linguistic findings further demonstrate modality-associated differences between the two groups. Participants in the ChatGPT condition produced higher levels of social processes, affiliation language, and self-referential pronouns, whereas participants in the MR condition demonstrated a higher emotional tone. These patterns are consistent with anthropomorphism theory, which proposes that

users respond socially to conversational agents through linguistic framing (Nass and Reeves 1996). Rather than indicating a singular effect of embodiment, the results suggest that different technological affordances may be associated with distinct configurations of embodiment and linguistic-anthropomorphic orientation. Therefore, these findings support the view that anthropomorphism is enacted interactively and may vary according to the semiotic resources made available within particular technological contexts (Seeger et al. 2021; Wood 2024). Accordingly, the affordances of embodiment must also be considered alongside accessibility and scalability. Mixed reality systems enable spatially integrated multimodal interactions but require specialized hardware and institutional resources. In contrast, voice-based conversational AI can be deployed through widely available personal devices, making it more scalable across educational contexts. These differences suggest that anthropomorphic engagement should be evaluated not only semiotically but also in light of pedagogical and infrastructural constraints.

Although educational performance outcomes were not the primary focus of this study, both conditions were embedded within a structured pedagogical framework and supported by human educator guidance. Across the six-week intervention, students demonstrated an increased frequency of interview practice and sustained engagement with the task. The MR condition's heightened emotional tone may have supported affective immersion and confidence-building, while the ChatGPT condition's elevated social-process language may have facilitated explicit rehearsal of relational and self-presentational discourse. Importantly, these gains occurred in conjunction with structured instructional scaffolding, suggesting that technological affordances alone do not determine educational effectiveness. Rather, the integration of AI-mediated interaction within a guided pedagogical framework appears central to meaningful learning outcomes.

6.1 Limitations and future research

This study has several limitations. First, the study was conducted within a single institutional and pedagogical context, which may limit its transferability to other disciplines or age groups. Second, while multimodal coding reached substantial interrater agreement (78 %), embodied interaction analysis inevitably involved interpretive judgment. Future research could incorporate eye-tracking or motion capture technologies to enhance behavioral precision. Third, the differences between MR and ChatGPT voices may reflect interface affordances as much as anthropomorphic perception. Comparative studies across additional embodied and voice-based platforms will clarify modality effects. Finally, although the intervention lasted six weeks, the HMD remains a comparatively novel stimulus relative to a

smartphone. As such, engagement levels in the MR group may partly reflect residual novelty effects, whereby initial curiosity or heightened attention toward advanced AI systems could have influenced behavioral indicators beyond what might be observed under fully stabilized usage conditions. (Miguel-Alonso et al. 2024).

The use of binary presence coding does not capture the frequency or duration of behaviors. Future research employing density-based measures (e.g., gestures per minute or sustained gaze duration) would provide a more fine-grained quantification of multimodal engagement.

Longitudinal research beyond six weeks could also examine whether anthropomorphic engagement stabilizes, intensifies, or diminishes over time, particularly as novelty effects fade and artificial intelligence (AI) literacy increases.

7 Conclusion

This study demonstrates that anthropomorphic perception in human–AI interaction is not merely a self-reported attitude but a multimodal and interactionally constructed phenomenon. By integrating multimodal social semiotics with LIWC-based linguistic analysis, the findings show that mixed reality elicited more embodied engagement, embodied engagement, and socially oriented language, whereas ChatGPT voice interactions elicited more linguistic engagement. These differences highlight how modality shapes meaning-making, social accountability, and learner positioning with respect to AI coaches. This study contributes a methodological model for aligning psychological theory with observable communicative practices in AI-mediated language learning environments.

References

- Abercrombie, Gavin, Amanda Cercas Curry, Tanvi Dinkar, Verena Rieser & Zeerak Talat. 2023. Mirages: On anthropomorphism in dialogue systems. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 4776–4790. Stroudsburg, PA: Association for Computational Linguistics.
- Ali, Duha, Yasin Fatemi, Elahe Boskabadi, Mohsen Nikfar, Jude Ugwuoke & Haneen Ali. 2024. ChatGPT in teaching and learning: A systematic review. *Education Sciences* 14(6). 643.
- Alkhayat, Amany Haroun. 2026. *The effect of multimodal conversational AI on job interview anxiety and performance among ESL students*. New York, NY: Teachers College, Columbia University dissertation.
- André, Elisabeth & Catherine Pelachaud. 2010. Interacting with embodied conversational agents. In Frank Chen & Kristiina Jokinen (eds.), *Speech technology: Theory and applications*, 123–149. New York: Springer US.
- Argyle, Michael, Mark Cook & Duncan Cramer. 1994. Gaze and mutual gaze. *The British Journal of Psychiatry* 165(6). 848–850.

- Avgustis, Iuliia. 2024. *Smartphone use by young adults: Collaborative practices of (dis-/re-)engagement in face-to-face interaction*. Oulu: University of Oulu dissertation.
- Bartneck, Christoph, Dana Kulić, Elizabeth Croft & Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics* 1(1). 71–81.
- Boyle, Elizabeth A., Anne H. Anderson & Alison Newlands. 1994. The effects of visibility on dialogue and performance in a cooperative problem solving task. *Language and Speech* 37(1). 1–20.
- Bozkurt, Aras. 2023. Unleashing the potential of generative AI, conversational agents and chatbots in educational praxis: A systematic review and bibliometric analysis of GenAI in education. *Open Praxis* 15(4). 261–270.
- Chen, Qian Qian & Hyun Jung Park. 2021. How anthropomorphism affects trust in intelligent personal assistants. *Industrial Management & Data Systems* 121(12). 2722–2737.
- Chen, Steven Szu-Chi & Henry Been-Lirn Duh. 2019. Interface of mixed reality: From the past to the future. *CCF Transactions on Pervasive Computing and Interaction* 1(1). 69–87.
- Chi, Michelene T. H., Stephanie A. Siler, Heisawn Jeong, Takashi Yamauchi & Robert G. Hausmann. 2001. Learning from human tutoring. *Cognitive Science* 25(4). 471–533.
- Chung, Cindy K. & James W. Pennebaker. 2012. Linguistic inquiry and word count (LIWC): Pronounced “Luke,” ... and other useful facts. In Paul M. McCarthy & Chutima Boonthum-Denecke (eds.), *Applied natural language processing: Identification, investigation and resolution*, 206–229. Hershey, PA: IGI Global Scientific Publishing.
- Cohen, Myke C., Mustafa Demir, Erin K. Chiou & Nancy J. Cooke. 2021. The dynamics of trust and verbal anthropomorphism in human-autonomy teaming. In *2021 IEEE 2nd International Conference on Human-Machine Systems (ICHMS)*, 1–6. Piscataway, NJ: IEEE.
- Cyr, Dianne, Khaled Hassanein, Milena Head & Alex Ivanov. 2007. The role of social presence in establishing loyalty in e-service environments. *Interacting with Computers* 19(1). 43–56.
- Epley, Nicholas, Scott Akalis, Adam Waytz & John T. Cacioppo. 2008. Creating social connection through inferential reproduction: Loneliness and perceived agency in gadgets, gods, and greyhounds. *Psychological Science* 19(2). 114–120.
- Epley, Nicholas, Adam Waytz & John T. Cacioppo. 2007. On seeing human: A three-factor theory of anthropomorphism. *Psychological Review* 114(4). 864–886.
- Gibson, James. 1979. *The ecological approach to visual perception*. Boston: Houghton Mifflin.
- Gray, Heather M., Kurt Gray & Daniel M. Wegner. 2007. Dimensions of mind perception. *Science* 315(5812). 619.
- Greilich, Angela, Kerstin Bremser & Kirsten Wüst. 2025. Consumer response to anthropomorphism of text-based AI chatbots: A systematic literature review and future research directions. *International Journal of Consumer Studies* 49(5). e70108.
- Gulay, Emreacan, Eleonora Picco, Enrico Glerean & Corinna Coupette. 2025. Relational dissonance in human-AI interactions: The case of knowledge work. arXiv:2509.15836.
- Iverson, Jana M. & Susan Goldin-Meadow. 1998. Why people gesture when they speak. *Nature* 396(6708). 228.
- Jeon, Jaeho & Seongyong Lee. 2023. Large language models in education: A focus on the complementary relationship between human teachers and ChatGPT. *Education and Information Technologies* 28(12). 15873–15892.
- Jin, S. Venus & Seounmi Youn. 2023. Social presence and imagery processing as predictors of chatbot continuance intention in human-AI interaction. *International Journal of Human-Computer Interaction* 39(9). 1874–1886.

- Jindapitak, Naratip & Munir Laeha. 2026. When the machine has a mind: How learners enact anthropomorphic scripts in AI-mediated interaction. SSRN 6056935.
- Kahn, Jeffrey H., Renee M. Tobin, Audra E. Massey & Jennifer A. Anderson. 2007. Measuring emotional expression with the Linguistic Inquiry and word count. *The American Journal of Psychology* 120(2). 263–286.
- Kaplan, Roza & Rusen Meylani. 2025. Dimensions of artificial intelligence literacy: A qualitative synthesis of contemporary research literature. *Journal of Computer and Education Research* 13(26). 790–825.
- Kendrick, Robin H., Judith Holler & Stephen C. Levinson. 2023. Turn-taking in human face-to-face interaction is multimodal: Gaze direction and manual gestures aid the coordination of turn transitions. *Philosophical Transactions of the Royal Society B: Biological Sciences* 378(1875). 20210473.
- Koch, Michael, Kai von Luck, Jan Schwarzer & Susanne Draheim. 2018. The novelty effect in large display deployments: Experiences and lessons-learned for evaluating prototypes. In *Proceedings of the 16th European Conference on Computer-Supported Cooperative Work: Exploratory Papers*. Siegen: European Society for Socially Embedded Technologies (EUSSET).
- Kress, Gunther. 2009. *Multimodality: A social semiotic approach to contemporary communication*. London: Routledge.
- Kress, Gunther & Theo Van Leeuwen. 2020. *Reading images: The grammar of visual design*. London: Routledge.
- Landis, J. Richard & Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics* 33(1). 159–174.
- Lee, Joomi, Crystal Chen Chen & Aryabrata Basu. 2025. From novelty to knowledge: A longitudinal investigation of the novelty effect on learning outcomes in virtual reality. *IEEE Transactions on Visualization and Computer Graphics* 31(5). 3204–3212.
- Li, Mengjun & Ayoung Suh. 2021. Machinelike or humanlike? A literature review of anthropomorphism in AI-enabled technology. In *Proceedings of the 54th Hawaii International Conference on System Sciences (HICSS 2021)*, 4053–4062.
- Liu, Zhengyuan, Stella Xin Yin, Carolyn Lee & Nancy F. Chen. 2024. Scaffolding language learning via multi-modal tutoring systems with pedagogical instructions. In *2024 IEEE Conference on Artificial Intelligence (CAI)*, 1258–1265. Piscataway, NJ: IEEE.
- Luckyardi, Senny, Juliana Karin, R. Rosmaladewi, Achmad Hufad & Nuria Haristiani. 2024. Chatbots as digital language tutors: Revolutionizing education through AI. *Indonesian Journal of Science and Technology* 9(3). 885–908.
- Miguel-Alonso, Ines, David Checa, Henar Guillen-Sanz & Andres Bustillo. 2024. Evaluation of the novelty effect in immersive virtual reality learning experiences. *Virtual Reality* 28(1). 27.
- Nass, Clifford & Youngme Moon. 2000. Machines and mindlessness: Social responses to computers. *Journal of Social Issues* 56(1). 81–103.
- Nass, Clifford & Byron Reeves. 1996. *The media equation: How people treat computers, television, and new media like real people and places*. Stanford: CSLI Publications.
- Pennebaker, James W., Ryan L. Boyd, Kayla Jordan & Kate Blackburn. 2015. *The development and psychometric properties of LIWC2015*. Austin: University of Texas at Austin.
- Rheu, Minjin, Ji Youn Shin, Wei Peng & Jina Huh-Yoo. 2021. Systematic review: Trust-building factors and implications for conversational agent design. *International Journal of Human-Computer Interaction* 37(1). 81–96.
- Salles, Arleen, Kathinka Evers & Michele Farisco. 2020. Anthropomorphism in AI. *AJOB Neuroscience* 11(2). 88–95.
- Schneider, Sascha, Steve Nebel Maik Beege & Günter Daniel Rey. 2018. Anthropomorphism in decorative pictures: Benefit or harm for learning? *Journal of Educational Psychology* 110(2). 218–232.

- Seeger, Anna-Maria, Jella Pfeiffer & Armin Heinzl. 2018. Designing anthropomorphic conversational agents: Development and empirical evaluation of a design framework. In *Proceedings of the 39th International Conference on Information Systems (ICIS 2018)*. San Francisco: Association for Information Systems.
- Seeger, Anna-Maria, Jella Pfeiffer & Armin Heinzl. 2021. Texting with humanlike conversational agents: Designing for anthropomorphism. *Journal of the Association for Information Systems* 22(4). 931–967.
- Slater, Mel. 2009. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B: Biological Sciences* 364(1535). 3549–3557.
- Spatola, Nicolas, Serena Marchesi & Agnieszka Wykowska. 2022. Different models of anthropomorphism across cultures and ontological limits in current frameworks: The integrative framework of anthropomorphism. *Frontiers in Robotics and AI* 9. 863319.
- van Lier, Leo. 2004. *The ecology and semiotics of language learning: A sociocultural perspective*. Dordrecht: Springer Netherlands.
- Wood, Geert. 2024. *Exploratory study: The impact of AI literacy training on perception of anthropomorphism and trust in conversational AI*. Eindhoven: Eindhoven University of Technology MA thesis.
- Zawacki-Richter, Olaf, Victoria I. Marín, Melissa Bond & Franziska Gouverneur. 2019. Systematic review of research on artificial intelligence applications in higher education: Where are the educators? *International Journal of Educational Technology in Higher Education* 16. 39.
- Zhang, Yongxiang & Weiwei Pan. 2025. A scoping review of embodied conversational agents in education: Trends and innovations from 2014 to 2024. *Interactive Learning Environments* 33(7). 4566–4587.

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